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Branding is what people say about you when you are not in the room.



-Jeff Bezos



LIVE DIY: LED Dimmer Circuit Using 555 Timer

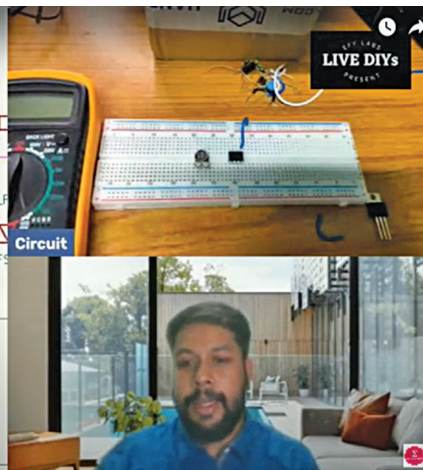
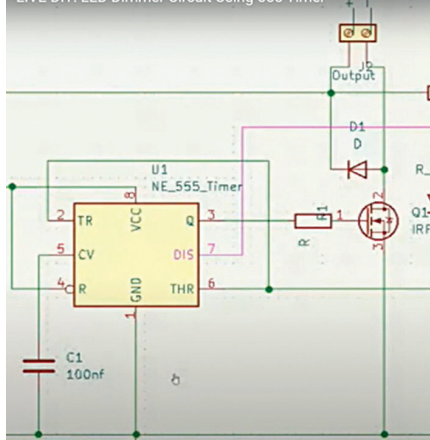


Fig. 7: Screenshot from recorded video of the project's demonstration

switch, which can be excited by the output of 555 timer and is suitable for the voltage requirement of our load. It has following specifications:

VDSS = 55V

RDS(on) = 17.5mΩ

ID = 49A

Controlling intensity of LED

As mentioned earlier, passive components in the resistor and capacitor (RC) network can be used to control the duty cycle and therefore the brightness of the LED.

$$\text{Duty cycle (D)} = \frac{\text{PW}}{\text{T}}$$

Here, D = Duty cycle, PW = Pulse Width (pulse active time), and T = Total period of the signal

Duty cycle represents the ratio of time when the load is connected to the power compared to the time load is not connected to the power. The higher duty cycle means that the load is connected to the power source for a longer duration compared to a lower duty cycle.

This technique can be used to control the intensity of LEDs or the speed of a motor. When the duty cycle is high, the LED glows brighter or the speed of the motor will be greater. To reduce the intensity of the LEDs, or to reduce the RPM of the motor, the duty cycle needs to be reduced.

$$\text{Intensity of LEDs} = \left(\frac{\text{On time}}{\text{total time}} \right) \times \text{max illumination}$$

$$\text{Speed} = \left(\frac{\text{On time}}{\text{total time}} \right) \times \text{max speed}$$

Testing

To check whether the circuit works properly or not, you may use a 12V battery pack (like the author) as input. Since we are using an N-channel MOSFET and providing a negative switching control, the positive terminal of the load is always connected while the negative terminal is connected to the drain of the MOSFET, which is driven by the 555 timer IC. Fig. 5 shows the LED strip glowing at the maximum brightness.

To reduce the LEDs' intensity, turn the knob of the potentiometer anticlockwise, which reduces the brightness of LEDs, as shown in Fig. 6.

The same setup may be tested with a 12V DC motor. We should be able to control the speed of the motor. A live demonstration of this circuit may be seen by watching the video at <https://urlcc.cc/dimmer> **EFY**

BONUS. You can watch the VIDEO of the tutorial of this DIY project at: <https://www.electronicsforu.com/videos/slideshows/live-diy-led-dimmer-circuit-using-555-timer>

The author, Sharad Bhowmick, works as a Technology Journalist at EFY. He is passionate about power electronics and energy storage technologies. He wants to help achieve the goal of a carbon neutral world.